

Positioning and focusing of light sources in light concentrating systems using convolutional neural network modelling

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ABSTRACT

Light energy concentrating systems such as High Flux Solar Simulators (HFSS) offer notable advantages in renewable energy research. Despite their expanding usage, they face some operational challenges that provide opportunities for new designs and further exploration. Accordingly, this study investigated the development and performance of a machine learning model based on convolutional neural networks (CNNs) for HFSS operation and control. Specifically, the hypothesis proposed is that a CNN can predict three HFSS parameters (the relative x , y , z position of the light source) using imaging and computer vision techniques with an accuracy equal to or better than the operator. First, the HFSS output was characterized in detail to set the overall study expectations and serve as the baseline metric. Then, the optical modelling of the HFSS using Monte Carlo Ray Tracing was employed to generate more than 2,500 images for the training and validation of the CNN. Consequently, the hypothesis was validated as the CNN accurately predicted the source position within 0.07, 0.11, and 0.07 mm in the x -, y -, and z -directions, respectively (a Euclidean distance of ~ 0.249 mm). This is comparable to the mechanical system's accuracy, which allows the positioning of the light source with a Euclidean distance of approximately 0.24 mm. Remarkably, this achievement allows for full automation and better control of light concentrating facilities and the development of more innovative energy harvesting systems.

1. Introduction

One of the crucial engineering challenges of this time is to provide a cleaner source of energy (Abbas and Wan Daud, 2010) to power rapid industrial and population growth. While significant research has been conducted on materials and energy harvesting concepts (Cheng et al., 2020), there is a need for advanced testing methods with quick turn-arounds and equipment with flexible control (Garg et al., 1985). Various research fields exploit concentrated sunlight for applications such as electricity production (Renno et al., 2017), chemical synthesis (Rego de Vasconcelos and Lavoie, 2019), and CO₂ utilization (Khan and Tahir, 2019; Nguyen et al., 2020). The need for pilot-scale and well-controlled experimental conditions led to the development and construction of light concentrating facilities referred to as high flux solar simulators (Gallo et al., 2017), ranging from a few kW_e to more than 300 kW_e (Wiegardt et al., 2017). These new apparatuses illuminate previously unknown challenges, such as the observer effect (Rowe et al., 2017), and bear some typical industrial issues, such as the design of a suitable control system (Martínez-Manuel et al., 2018). Simultaneously, they

offer opportunities to explore complex control strategies (Rowe et al., 2018), material testing methods (Sarwar et al., 2018), optimization of new devices (Gadi, 2000; Sarwar et al., 2019), novel reactor design (Gu et al., 2019), and more. We explored one such opportunity here: using machine learning algorithms to position lamps within a light concentrating facility.

High flux solar simulators (HFSS) typically consist of multiple light sources coupled with ellipsoidal or parabolic reflectors (Dibowski, Sarwar et al., 2015b). One advantage of a HFSS is the ability to concentrate the light to regions of different areas on the focal plane with Gaussian (Zhu et al., 2020) or top-hat (Jin et al., 2019) profiles. This means it is possible to control and optimize the incident flux on equipment such as high temperature reactors (Guene Lougou et al., 2020; Sarwar et al., 2015a,b,c) and photovoltaic cells (Kang et al., 2018) placed on the focal plane. Several multi-lamp HFSSs were recently reviewed by Gallo et al. (2017), where the importance of facility flux characterization was emphasized. One method of characterization is collecting images of the illuminated area, correlating pixel values with actual measured irradiance levels, and reconstructing irradiance distribution (Neumann and

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Groer, 1996). Moreover, during the characterization process, the operator must fine-tune three to six parameters (degrees of freedom - e.g., x-y-z position and roll-pitch-tilt angles) per light source with very high accuracy. Even submillimeter changes during the tuning process affects the system's performance (Gallo et al., 2017; Rowe et al., 2018; Sarwar et al., 2015a). Even though this is a tedious and demanding process, it is mostly a manual process that is confined to flux maximization and uniformity optimization in irradiance imagery (Levêque et al., 2016). This may be because flux characterization produces just a series of monochromatic images and contour plots of the irradiance and flux (flux maps). While there are tools such as those based on machine learning that can utilize these images, they have not yet been tested for this purpose. The automation of this process through machine learning may allow for tailored flux profiles that match or achieve the distribution of sunlight in field applications.

Machine learning uses statistical models and methods to inform a mathematical model that relates the inputs and outputs. The application of machine learning techniques to current research topics has advanced research at a remarkable rate. Examples include autonomous equipment operation (Bishop, 2006), hydrogen production optimization (Ren et al., 2020), adaptive engine control (Wong and Wong, 2018), and prediction of future energy trends (Cheng et al., 2020). Because of the field's immense depth, an interested reader can learn more details in any of the most recent state-of-the-art analyses, such as Abiodun et al. (2018). One of the standard applications of machine learning is image classification (Schmidhuber, 2015). Convolutional neural networks (CNNs) are a sub-classification of a broader family of machine learning methods called *deep learning*. CNNs are extensively used to relate image features such as edges and colors with data such as a continuous or discrete variable. The establishment of such a relationship is an iterative approach known as training, which requires large datasets and tuning of multiple hyper-parameters (Goodfellow et al., 2016; LeCun et al., 2015).

In this study, we investigated the development and performance of a CNN on the operation and control of a HFSS light concentrating facility. We utilized images of a target illuminated by the HFSS, both captured by a camera and generated by optical modelling, to train a CNN to predict the light source's relative position. Our hypothesis was that CNNs can predict the light source's position with accuracy equal to or better than the human operator. Proving this hypothesis will allow for faster configuration and optimization of any multi-lamp HFSS. It will also offer the capacity to generate flux maps (irradiances) beyond the typical form (i.e., concentric contours with Gaussian distribution of flux) and power innovative and radical light concentrating technologies.

2. Methodology

This section elaborates on the four main methods employed in this study. It starts with details of the experimental setup (i.e., HFSS at Texas A&M at Qatar), which also sets the quantitative metric to be achieved, i.e., the accuracy of a human operator to position the light source. It continues with the description and the outcomes of the HFSS's flux characterization, including the post-processing of the captured images. The optical modelling outline is then provided, including the virtual data generation approaches, but without an in-depth description of the optical modelling software. Finally, it concludes with the development steps of the CNN and illustrative examples of selected hyper-parameters.

2.1. Experimental setup

The HFSS (Fig. 1) consisted of seven independent lamp-blocks, each equipped with a truncated ellipsoidal and aluminum coated reflector, a 6 kW Xe short-arc ozone-free lamp (by Osram), and the necessary power supply electronics and air-handling units. The six lamp-blocks were placed on a *squeezed* hexagonal configuration with the seventh block in the center. The HFSS was enclosed in a $7 \times 3.5 \times 2.5$ m³ (LxWxH) container made with prefabricated aluminum composite panels for increased user safety, reduction of light bleeding, and temperature control. The HFSS included the target area located at a focal distance of 2.97 m. It was also equipped with two smoke and one flammable-gas detector, and two ceiling openings to allow the removal of potential fumes and gas leaks. The overall design of the HFSS was similar to most such systems (e.g., University of Minnesota, Georgia Tech, University of Florida, and ETH Zurich) and especially that of Lorentzou et al. (2017) at CErTH-Greece. One additional element is the remote-controlled manipulation of the lamp position (3-axis), which allowed for accurate positioning of the lamp (arc/ filament). To our knowledge, this is a rare configuration for a HFSS. Some HFSSs with this feature are present in the University of Colorado (US) and CErTH (Greece). The lamp position was controlled using a slider developed in-house that used a 56 mm slider-type potentiometer per axis to monitor the position and a microcontroller with a 10-bit analog-to-digital converter (ADC) to operate the slider. The calculated resolution was 0.06 mm (using the slider length and ADC resolution). However, upon measurement with a precision micrometer caliper, the actual resolution was determined to be around 0.14 mm. The latter was assumed as the maximum accuracy (minimum uncertainty) of a human operator to position the light source at each direction; alternatively defined as a Euclidean distance, $|d|_{\text{HFSS}}$, of 0.24 mm ($|d| = \sqrt{3 \cdot 0.14^2}$).

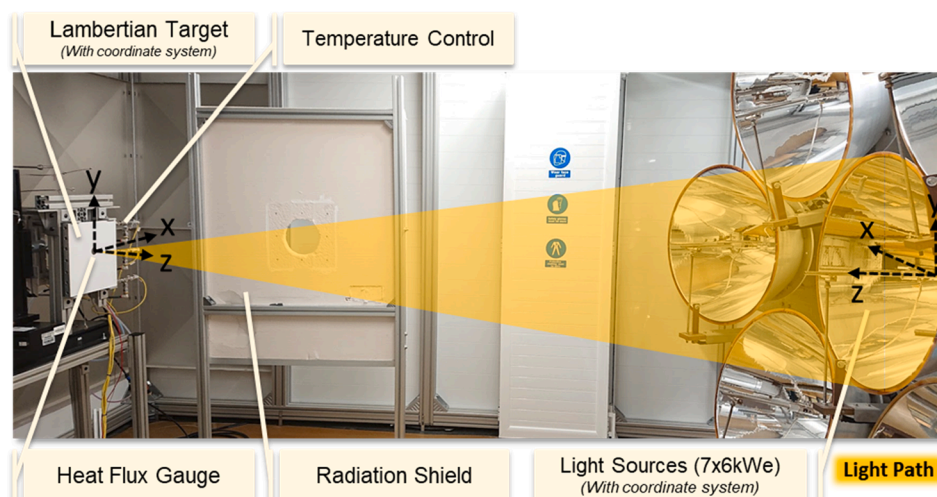


Fig. 1. Photo of the HFSS facility's interior and its main components with a graphic representation of the central light source's light path and the coordinate systems for the Lambertian target and the light source.

At the target area, a Lambertian reflector (a 250x250 mm² stainless steel slab plasma coated with Al₂O₃; made to order by Haueter Engineering, Switzerland) was mounted on a 3-axis slider (NLS8–500–101, NLS8–300–102, and NLS8–200–101 precision sliders by NewMark) with a resolution of 0.08 µm and a 1 µm repeatability. The Lambertian reflector ensured the same apparent brightness irrespective of the observer's viewing angle (Shimokawa et al., 2001). A monochrome camera with an 8-bit and 1/1.8" charged-coupled device (CCD) sensor (1600x1200 DMK 51BG.02.H by TheImagingSource) with 75 mm lens (C7528-M by Pentax) was used to capture images of the Lambertian target. It was placed 50 cm above the central lamp and equipped with one neutral density filter (Optical Density 4.8 by Aulektron). A Gardon-type circular foil radiometer-heat flux gauge (Vatell TG1000-0) with a colloidal graphite coating was installed at the Lambertian target's center. The heat flux gauge was recently calibrated under a high-intensity lamp against a traceable NIST sensor (by Hukseflux, with 1% repeatability, 2 mV.(W.cm⁻²)⁻¹ sensitivity, and transducer calibration accuracy of ± 3%). The later sensor was calibrated under black body radiation. It was important to modify the calibration factor using *calorimetric calibration* (Krueger et al., 2013) to account for the differences in the radiation sources' spectra and the spectral absorptivity of the radiometer coating. Details of the calorimetric calibration method related to this study have been reported earlier (Sarwar et al., 2015a; Sarwar et al., 2014; Sarwar et al., 2017). The heat flux gauge output on the order of 0–10 mV was logged using a 24-bit ADC (cRIO module NI-9211 by National Instruments) with 0.008 mV sensitivity. The Lambertian target and flux gauge were both water-cooled at 25 degreesC (using an F50-ME Julabo chiller) and equipped with multiple K-type thermocouples to monitor their temperature and prevent overheating.

2.2. Flux characterization

The mapping of the irradiance on the target is a three-step method known as flux characterization (Gallo et al., 2017; Levêque et al., 2016), which allows for the determination of the flux's spatial distribution with high accuracy and minimum direct measurements. In principle, the flux characterization method correlates the radiative incident flux values to the grayscale values of the target photos. The grayscale value represents the brightness of a pixel in a monochromatic image obtained from the CCD camera. After characterization, each pixel value corresponds to a specific incident radiation flux value, hence reconstructing the flux's spatial distribution from the target photos. The following paragraphs describe in detail the three steps and the characterization results for the central lamp.

At first, flux measurements across the radiation focus area were collected automatically using in-house software (developed in LabVIEW) to control the 3-axis slider. In this study, the flux measurements were collected in three intervals in the x- and y-axis (see Fig. 1 for the coordinate system), i.e., every 0.25 mm for the first 10 mm from the center, every 0.5 mm for the next 20 mm, and every 2 mm for the next 30 mm. These intervals were chosen to account for the rapid flux reduction moving outwards from the center, i.e., the typical Gaussian form of the focused flux. The data collection was applied to the negative x-direction, positive y-direction, and the diagonal of positive x-direction & negative y-direction. Furthermore, measurements were collected for the positive-x and negative-y directions to check for spatial non-uniformity, but this was not used in the characterization. The radiometer stabilized for 60 s at every new position, and then 30 flux measurements (rate 1 Hz) were collected and averaged. Thirty data points were considered sufficient to minimize standard deviation close to the sensitivity of the radiometer.

In the second step, photos of the Lambertian target were collected using the CCD camera at five different exposures: 1/500 s, 1/833 s, 1/1250 s, 1/2000 s, and 1/3333 s. To account for and remove the noise introduced by the CCD sensor, 40 photos were taken, averaged at each exposure level, and smoothed using a 2D-Gaussian kernel with a

standard deviation equal to 9 (empirically selected). The dark current signal, originating from the CCD sensor, was captured by collecting photos at each exposure level with all lamps extinguished and the enclosure doors closed, but with the target viewable through the camera filters. This dark current signal was subtracted from the averaged images. The final single exposure images (Fig. 2A) were constructed using a projective transformation algorithm and a rigidly affixed 6x4 chess-board reference paper (in-house, 40x40 mm² squares). This correction is routine but essential because of the inherent lens distortion and the camera's oblique point-of-view; interested readers can access more details on the related implications in Gill et al. (2015). Each of these single exposure images can store only a portion of the actual light intensity. For example, the high exposure photo (1/500 s) can store the low intensities but gets oversaturated at the large intensities. The opposite is true for the lowest exposure photos (1/3333 s). Combining photos at multiple exposures using High-Dynamic-Range and SuperResolution (HDR-SR) processing algorithms (Banterle et al., 2011) allows for the construction of an improved image with most information retained and adequately represented (Fig. 2B). For the same reason, we selected the exposures enumerated above to guide operations between 10% and 90% of the 12-bit grayscale (monochromatic) image for the selected target area (45 mm radius).

In addition, another averaging approach was tested by normalizing each image with its respective exposure as in the equation below:

$$\text{Normalized image}(i,j) = \sum_{n=1}^N \frac{P_n(i,j)}{t_{\text{expo},n}} \quad (1)$$

where $P_n(i,j)$ is the grayscale value $\in [0,1]$ of the n^{th} image's (i,j) pixel, and $t_{\text{expo},n}$ [s⁻¹] is the exposure of that image. Comparison of the two methods yielded similar grayscale vs. irradiance correlations, so we decided to continue with the former method, which allowed for upscaling of the image to 12-bit in a single step. However, the latter method was significantly faster (greater than 3x).

The combination of the flux measurements (step 1) and the HDR-SR image (step 2) was the third step, which resulted in the chart of Irradiance [W.m⁻²] vs Normalized Image Irradiance [0,1] (Fig. 3A) as well as two correlation equations: a linear relationship and a power-law relationship. The equations' constants were obtained with the Least Squares Method using the *backslash* operator and the *fit* function in MATLAB. The goodness of fit was assessed with multiple metrics. The NRMSE (Normalized Root Mean Square Error – equal to zero for ideal fit) was approximately 0.86 for the linear correlation (with slope $a = 1186.4$ [W.m⁻²]) and 0.89 for the power-law correlation (with constant $a = 1117.5$ [W.m⁻²] and exponent $b = 0.903$). Use of either equations has been reported in the literature, with the linear one being more common (Neumann and Groer, 1996), especially for higher heat fluxes. Nevertheless, the power-law equation was selected for the rest of the study since it yielded better agreement. The radiometer (heat flux gauge) provides the total incident flux over its exposed area (radius 12.6 mm) and not point measurements. Consequently, the same pattern was applied to collect the Normalized Image Irradiance (average grayscale values) for the radiometer's projected position on the focal plane. Experimental data in Fig. 3A show a tail at high irradiance levels attributed to malfunctioning of the control system. Although this experimental error was easily identified, others may not be. With this in mind, we included these erroneous data points into our calculations to show that they only have a minor contribution to the outcome of the irradiance calibration for large datasets.

As mentioned earlier, the final outcome of the flux characterization was a high-resolution flux map (Fig. 3B) along with post-processing results, such as the horizontal and vertical profiles of the flux distribution (symbol plots in Fig. 3C). The last figure includes the approximated Gaussian distributions (solid line plots) for each of the profiles that adequately validate (NRMSE for x-axis ~ 0.07 and y-axis ~ 0.12) the hypothesis of the Gaussian form of the focused light. Indirectly, this

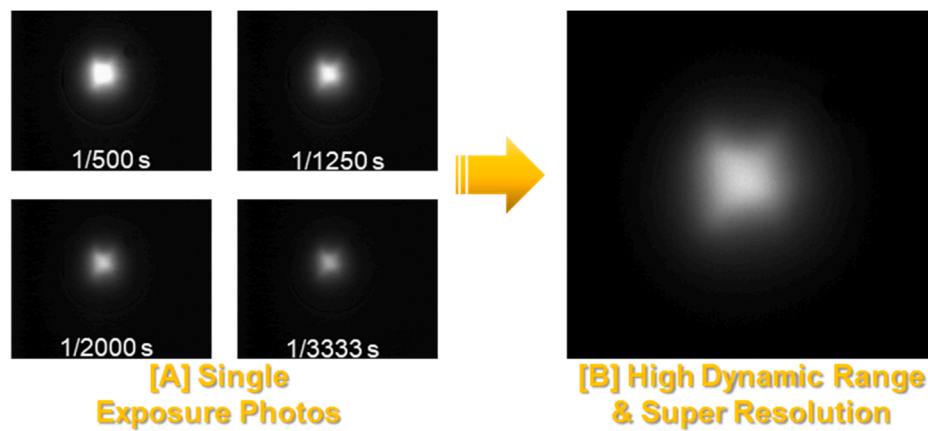


Fig. 2. Examples of A) single-exposure photos at four different exposure levels (1/800 s omitted for clarity) and B) the constructed high dynamic range and super-resolution image from the flux characterization method.

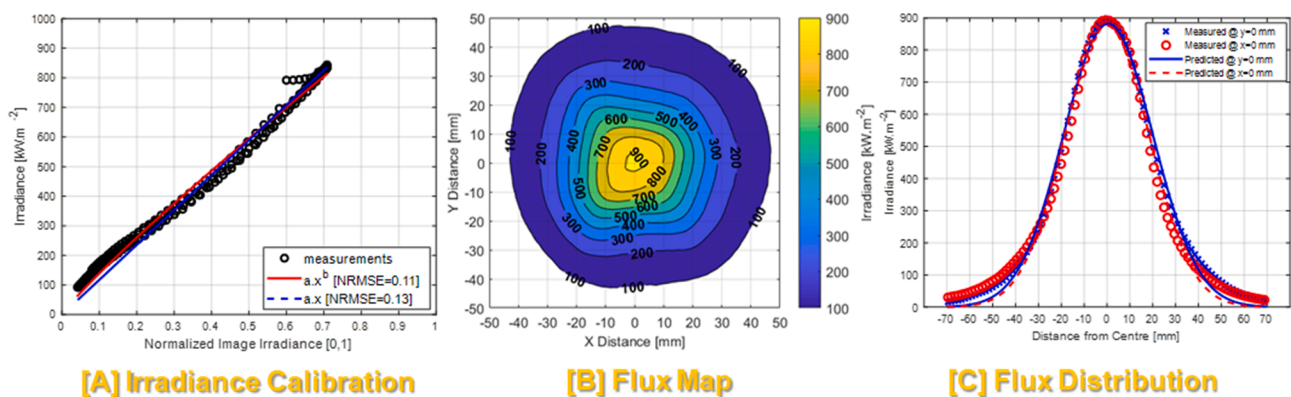


Fig. 3. Results of the flux characterization method: A) The irradiance calibration curve including a power-law plot with $a = 1117.5 \text{ [W.m}^{-2}]$ and $b = 0.903$, and a linear plot with $a = 1186.4 \text{ [W.m}^{-2}]$ where NRMSE is the Normalized Root Mean Square Error; C) the flux map based on the Fig. 2B; and C) the flux profiles across the x- and y- directions of the measured/estimated (plots with symbols) and predicted (line plots of fitted Gaussian distributions).

comparison verified the positioning of the reflector and the whole lamp block. If the lamp block were not correctly aligned with the target, the distribution would be skewed or asymmetric within the limits of reflector curvature and arc axial distortion.

All discussed image-processing algorithms were implemented in-house (using MATLAB, 2019) apart from the HDR-SR described in (Banterle et al., 2011), which is available as an external toolbox.

2.3. Optical modelling

The virtual flux maps were constructed with optical modelling using commercial ray-tracing software (TracePro 2018 by Lambda Research Corporation). In the literature, there is a significant number of such numerical tools, both open-source and commercial. For example, SolTrace is open-source software developed by the National Renewable Energy Laboratory (NREL) to model solar power concentrating systems, especially those with complex geometries using Monte Carlo ray tracing (MCRT). CUtrace is another Monte Carlo parallel ray tracer program available on MATLAB Central that is used for radiation analysis and view factor determination, especially for solar simulators (Rowe et al., 2015). Similarly, other software available includes Tonatiuh, OptiCAD, STRAL, and SPRAY, which have been reviewed by Osório et al. (2016). In any case, optical modeling allows for rapid acquisition of the (virtual) flux maps necessary in the training and testing phases of the neural network model development when compared to the tedious obtained experimental data.

The components of the central (see Fig. 1) lamp block (i.e., reflector

and lamp) were modelled using the built-in design tools of the software. The as-built Computer-Aided Design (CAD) 3D model of the lamp was also tested, yielding similar results, i.e., flux maps. However, the CAD model was overly complicated, comprising millions of triangular elements, which substantially slowed down the numerical simulations (approximately 20x). The representation of the arc (source of light) was an additional challenge with various approximations being suggested and tested, e.g., spherical, cylindrical (Costandy et al., 2012), probabilistic basis functions (Levêque et al., 2016), and combinations (Rowe et al., 2017). In the past, we employed a detailed representation of the arc using a combination of curved surfaces that mimic the brilliance distribution provided by the manufacturer (OSRAM, 2008) with very good agreement with the experimental flux map (Sarwar et al., 2015a; Sarwar et al., 2017). According to the same manufacturer's manual, the plasma ball alone accounts for approximately half of the light produced in the arc. Therefore, a simpler representation may be capable of adequately reproducing the light source with notably reduced computational times, which is one of this study's requirements, since it aims to produce thousands of flux maps. This turned out to be correct, as we were able to approximate the light source with a single sphere of 2.5 mm radius and an equally good statistical agreement (NRMSE less than 0.10) compared to our earlier work (Sarwar et al., 2015a). The spherical light source was placed at the first focal point of the ellipsoidal reflector (90 mm from the base), which was also taken as the origin of the Cartesian coordinates employed hereafter (Fig. 1). The light source had a flux of 6 kW (arbitrarily selected value), and it was arbitrarily configured as a Lambertian source of light with a wavelength of 500 nm. Specular

reflectance of 0.92 was assumed at the reflector's inner surface. The quartz glass arc enclosure was considered a semitransparent gray medium with an absorptivity of 0.1 (Petrasch et al., 2007).

The optical modelling software was programmed (using the proprietary language *Scheme*) to collect flux maps on the target at varying light source positions. An example is shown in Fig. 4A. Four different flux maps (dataset) generation/collection approaches were tested:

- Uniform variation of the light source position by moving it relative to the origin in 0.25 mm intervals for 10 mm across all directions and their combinations, yielding around 512,000 flux maps.
- Use of the same approach as (i) but with combinations limited to the four main diagonals (e.g., $\varphi = \theta = 45$ degrees in spherical coordinates), yielding around 556 flux maps.
- Use of varying intervals following a Gaussian distribution with a mean of zero and standard deviation of five while limiting the combinations as in (ii) yielding around 721 flux maps.
- Random sampling of 2,530 of flux maps from coordinates belonging to $[0, +30]$ mm or $[-30, 0]$ mm.

2.4. Development of the convolutional neural network model

Convolutional neural networks are popular choices for image processing, particularly classification problems, in deep learning (Sultana et al., 2020). When an image is input to a convolutional layer as a tensor, the layer processes the tensor a few elements at a time. For instance, if the kernel size is set to 5x5, the convolutional layer will process a five-by-five tensor of elements, or pixels. This processing is a simple element-wise multiplication and addition, which produces an output layer (Castro et al., 2020). This output layer is a new tensor with a weighting factor representing each kernel across the entire tensor. In this sense, passing an image through a convolutional layer allows the algorithm to examine the image in locally connected pockets, rather than an individual pixel at a time. This method is used in image processing because it is adept at discovering features, such as circles or lines with distinctive value gradients (Dhillon and Verma, 2020). HFSS flux maps most often are characterized by features such as concentric circles or arcs of different flux intensities, as seen in Fig. 4. The diameter and distribution of these features reveal significant clues to the focus of the lamp and are utilized by the operators. For this reason, convolutional networks were deemed a good fit for the objective of this study: to design a machine learning algorithm to predict the light source coordinates using a known flux map distribution.

The neural network was built in the Python (v3.7) programming language using Keras (v2.2.5) convolutional models with a TensorFlow backend (v1.13.2). After data was collected, separated into testing and training data, and normalized, a sequential Keras model was created. The network was compiled with the $loss = mean_squared_error$ metric as

it was a regressive network as opposed to a categorical one. The model's final layer was a fully connected (dense) layer with three output values, which would be trained against the three coordinates (x, y, z). At first, and with some adjustments, the training was able to reduce the model's loss to approximately 10^{-7} . However, further evaluation demonstrated low accuracy outside of the training data, revealing a large amount of overfitting in the model. A callback checkpoint was added to monitor the validation loss and save the weights and biases when the validation loss was at a minimum, indicating overfitting (i.e., $loss \ll validation\ loss$).

As part of developing the machine learning model, several vital decisions had to be made, such as the type of machine learning method, the optimization method, and the value of various hyperparameters (some examples presented in Fig. 5). The methodology behind these decisions is explained in more detail below:

- RMSE (Root Mean Square Error) is the overall representation of the quality of a regression line fit to the data and thus is a measure of the overall error function. RMSE is generally referred to as *loss* and may be used as an indicator of model precision. The validation loss is another metric to protect the algorithm from detecting patterns too limited and specific to the training data. Validation loss is also an RMSE calculation performed on a reserved validation data set not directly used for training. The effect of the hyperparameters was considered to be independent (Bengio, 2012).
- The *learning rate* can either be constant or drop slowly throughout the process (adaptive approach). The latter is preferred because it decreases computational time. The initial learning rate, drop factor, and drop period need to be configured when using the adaptive method. The initial learning rate is the most significant factor because if it is too large, the training will never converge, and if it is too small, the training will converge slowly. Because of this, this parameter was determined through trial and error with a typical range between 10^{-6} and 1. The drop factor reduces the learning rate periodically as per the specified drop period. This factor is not as crucial since if the learning rate drops too fast it will slow down the process without affecting the accuracy. A powerful system would allow for a larger drop rate at smaller drop intervals and vice versa (Brownlee, 2019).
- A *momentum* value of 0.9 was used based on recommendations from other research (Goodfellow et al., 2016). Since this factor's effect is weaker than the other factors, it was not investigated.
- Different kernel sizes were tested on the optical modelling data to determine the optimal size. It was found that a kernel size of larger than 10-by-10 pixels resulted in significant distortion on the first layer (e.g., Fig. 5B). Additionally, kernels of this size significantly increased computational time. It was found that maximum loss was achieved by stacking three convolutional layers with a kernel size of 2-by-2.

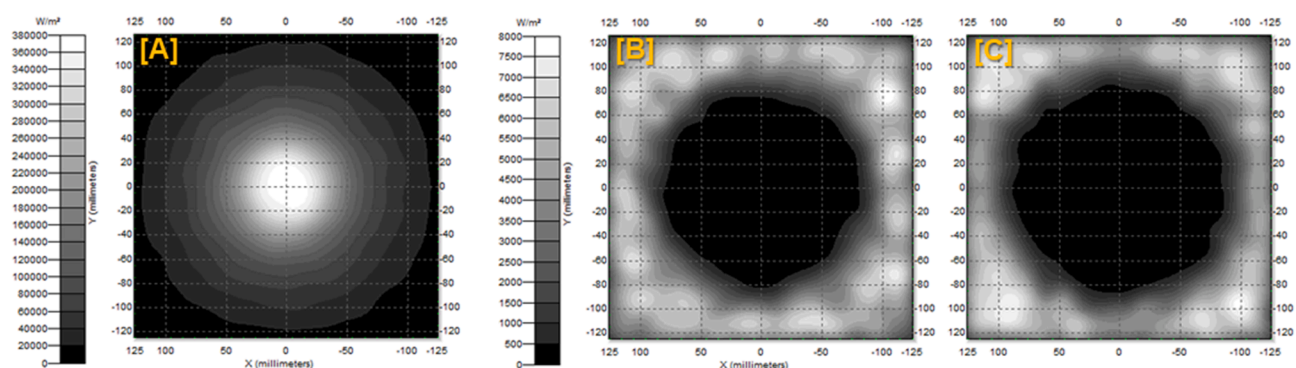


Fig. 4. Indicative flux maps (with an arbitrary selected source power of 6 kW) generated using the optical modelling software (TracePro®) with the light source positioned at the focal point (A); and shifted in the z-direction by -30 mm (B) and $+30$ mm (C).

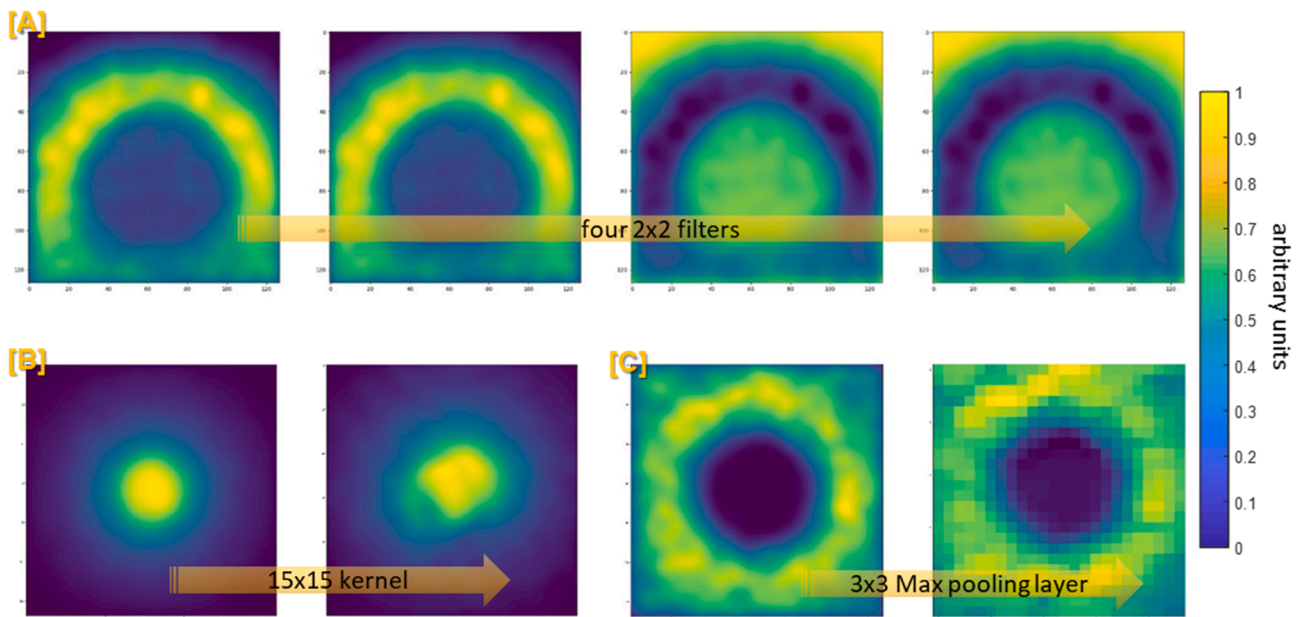


Fig. 5. Illustrative examples of the effects on a collected flux map (128x128 pixels) of A) four sequential filters (2x2 pixels), B) 15x15 convolutional kernel, and C) a Max pooling layer size 3x3 pixels. The color scale has arbitrary units and reflects values of the CNN internal weighing factors.

- Another significant parameter was the number of filters (e.g., Fig. 5A). Intuitively, this can be understood as the number of feature detectors or the number of patterns the convolutional layer can detect. For instance, in RGB (red/blue/green) image processing, if four filters are used, one may detect features in the red values, one in the blues, one in the greens, and perhaps one with a combination. In monochromatic flux maps, increasing the number of filters could increase the ability to detect more concentric circles, representing contour lines of flux.
- After the convolutional layer, a pooling layer was added. Pooling layers, like convolutional layers, sample the image piece by piece with a tensor of a given size. However, these layers simply downsize the scale of the image rather than extracting features. Two pooling types were tested: max pooling, which replaces that selection of pixels with a single pixel of their maximum value (example in Fig. 5C), and average pooling, which replaces the pixels with their average value. This type of downscaling was used to prevent the algorithm from learning small features that may not be important to the overall prediction.

3. Results and discussion

3.1. Data collection approaches

The first two data collection approaches that relied on fixed steps in the position of the light source proved problematic. The uniform variation (i) across all directions created a high number of flux maps, requiring multiple days of computation using optical modelling software. This approach ended up overfitting the model (see discussion on loss and validation loss in 2.4). In comparison, the generation of the flux maps (ii) by limiting the light source's position along each axis and the diagonals with fixed increments took only a few hours. However, the created model was only accurate when tested along those axes, but not for all of 3D space. The third (iii) collection approach was based on earlier observations on a different system employing a thermochemical reactor with a secondary concentrator (Khalil et al., 2019; Mohammed et al., 2019). In that system, movements of equivalent magnitude had a more substantial effect on the reactor temperature when the source was closer to the origin (focal point). Both experimental and numerical tests

invalidated this assumption because small light source movements made more significant changes to the flux map (total power and peak flux) when the lamp was already largely defocused.

The main findings from extensive tests with the first three collection approaches were that the data must represent all possible input data without using specific input patterns (e.g., fixed steps, Gaussian distribution steps) to ensure the algorithm does not learn these input patterns. These findings, along with other smaller observations, led to the fourth (iv) data collection approach: use of random number-generated co-ordinates within $[-15, +15]$ mm or $[-30, +30]$ mm being adopted. Contrary to our initial expectation, all tested combinations of filters, layers, and kernel sizes failed to achieve accuracy better than 0.8 mm in the x- and y-direction and 3 mm in the z-direction. There was little improvement despite incorporating multiple methods to prevent overfitting and small alterations in the data generation approach. A more careful observation of the input data revealed a prominent symmetry across the xy-plane at the origin point (z-direction). The flux maps generated with the optical modelling software were similar for symmetric distances along the z-direction (see Fig. 4B and C). This similarity was far less apparent on the experimental flux maps, probably due to the reflector's surface defects and other manufacturing imperfections. The CNN model was trained using either positive or negative z-axis positions only instead of utilizing both. This adjustment immediately showed significant improvement in the loss values, with less overfitting (demonstrated by similar loss and validation loss values) and higher accuracy. After this was discovered, all results utilized this adjustment.

More variations of the fourth (iv) data collection approach were tested. Among them, the training of the model on random data within $[0, +15]$ mm or $[0, +30]$ mm but with higher priority (more training data) afforded to the first 10 mm showed the most promising results. This finding strengthens the earlier point that the input data must represent the actual lab/real applications.

3.2. Evaluation of the convolutional neural network model

Finally, a dataset of 2,530 flux maps was generated using the fourth (iv) approach. Following recommended practices (Brownlee, 2019), a random sample of 2,155 was used to train the model. Of the remaining maps, 240 were held out to validate the model and provide robustness

against overfitting, and 135 were used to test the model. Overall, the achieved accuracy (Table 1) in each dimension was relatively similar: 0.163 mm in the x-direction, 0.141 mm in the y-direction, and 0.125 mm in the z-direction. These estimated accuracies/errors (CNN versus MCRT, see Fig. 6) can be combined into one absolute error represented by the Euclidean distance from the origin point, equal to $|d|_{\text{CNN}} = 0.249$ mm. This absolute error was within the acceptable range, given that the accuracy of the mechanical slider, which controls the position of the light source, was around 0.14 mm (see paragraph 2.1). Similarly, the absolute error introduced by the light-source mechanical slider was around $|d|_{\text{HFSS}} = 0.24$ mm.

The CNN model was fed equipment flux imagery to produce the predictions shown in Fig. 7. Approximately 75% of the light source positions of the (iv) approach was replicated using the experimental setup since some of the coordinates could not be achieved because of the mechanical slider's accuracy. The agreement was excellent for the data within the trained range of the CNN model but poorer for out-of-sample data.

While the machine learning model was designed to minimize the error in coordinate prediction, the real error inflicted on the system was the flux difference at each point. Accordingly, in addition to the above accuracy estimations, a series of error graphs were produced to visually inspect the differences between the numerical and predicted flux maps. Particularly, these graphs present the % error on the flux map using point-by-point subtraction of the flux maps produced via optical modelling (MCRT) for the original coordinates (as defined in the test sample) and the predicted coordinates (obtained by the CNN model). Two distinct comparison examples are presented in Fig. 8 and Fig. 9.

Fig. 8 presents an example where the deviation in all three dimensions (i.e., 0.17 mm in the x, 0.21 mm in the y, and 0.19 mm in the z) is larger than the estimated accuracy. The input flux map (Fig. 8A) with original light source coordinates of (-0.76, -7.14, 0.13) mm was fed into the CNN, which predicted the coordinates as (-0.59, -6.93, 0.32) mm, corresponding to a slightly different flux map (Fig. 8B). The error was most significant in the y-direction (Fig. 8C), locally reaching values higher than 72%. Nevertheless, the differences in the maximum and average fluxes were very low. At the original coordinates, the maximum flux was $189,909 \text{ W.m}^{-2}$ and the average flux was $49,615 \text{ W.m}^{-2}$, while at the predicted coordinates, they were $189,340 \text{ W.m}^{-2}$ and $50,023 \text{ W.m}^{-2}$ respectively, yielding an agreement of around 99.7% and 99.2%, respectively. The total incident fluxes were also similar, with an error of less than 1.8%.

Fig. 9 illustrates another example where only the z-direction presented a deviation worse than the estimated accuracy (i.e., 0.08 mm in the x, 0.04 mm in the y, and 0.22 mm in the z). The CNN model was provided the input flux map (Fig. 9A) whose original light source coordinates were at (-1.14, -4.94, 6.27) mm. The CNN model predicted the coordinates as (-1.06, -4.90, 6.49) mm, which corresponds to quite a different flux map (Fig. 9B). Although the local error values were less than 36% (Fig. 9C), they were widely spread throughout the map. As a result, the differences in the maximum, average, and total fluxes were larger than in the previous example. At the original coordinates, the

Table 1

Summary of the sample sizes used for the CNN model and the achieved accuracies compared to MCRT simulations.

Sample Size		
Training	Validation	Testing
2,155	240	135
Dimension		
Average Error [mm]		
x	0.163	
y	0.141	
z	0.125	
Euclidean Distance	0.249	

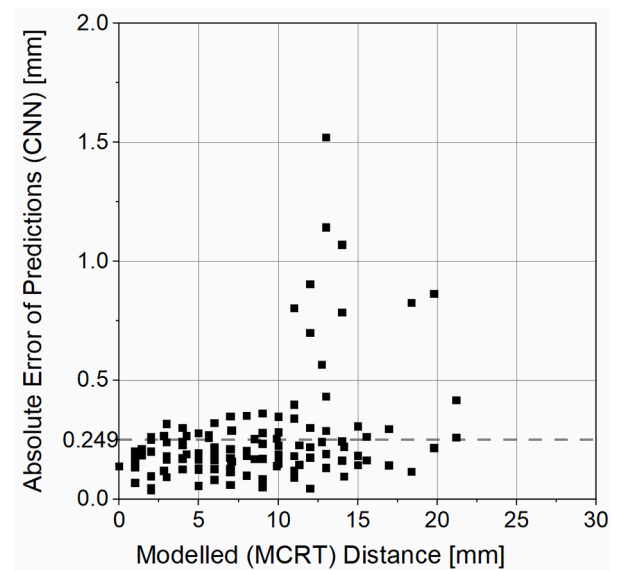


Fig. 6. Distribution of the absolute error as Euclidean distance of the modelled (using MCRT) coordinates from the predicted ones (using the CNN model) versus the distance of the modelled coordinates from the origin. The reference line at 0.249 mm represents the estimated absolute error (Euclidean distance, $|d|$) based on the individual x-, y-, and z-directions estimated accuracies.

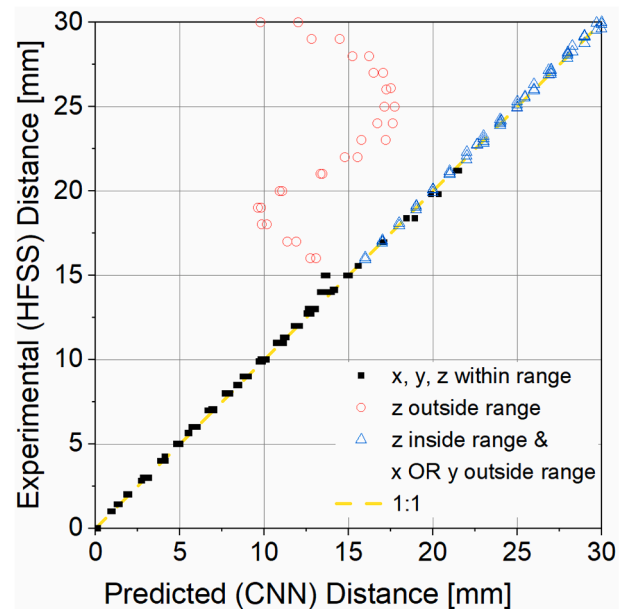


Fig. 7. Scatter plot of the Euclidean distances (from the origin) of the light source (experimental coordinates using HFSS) against the CNN predicted positions of actual equipment flux imagery. The solid symbols (black squares) represent the flux maps within the training range for x, y, z $\in [-15, 15]$ mm. The open symbols include the flux maps outside the training range, i.e., only for z $\in (15, 30]$ OR $[-30, -15]$ mm (red circles); and for both x & y $\in (15, 30]$ OR $[-30, -15]$ mm (blue triangles) while z is within the range. The diagonal line (dashed yellow) shows the 1:1 ratio. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

maximum flux was $153,364 \text{ W.m}^{-2}$ and the average flux was $43,942 \text{ W.m}^{-2}$. In comparison, at the predicted coordinates, these values were $148,674 \text{ W.m}^{-2}$ and $43,142 \text{ W.m}^{-2}$ respectively. This yielded slightly lower agreement than in the previous example, of around 96.9% and 98.2% respectively. In contrast, the total incident fluxes were still similar, with an error of less than 1.8%.

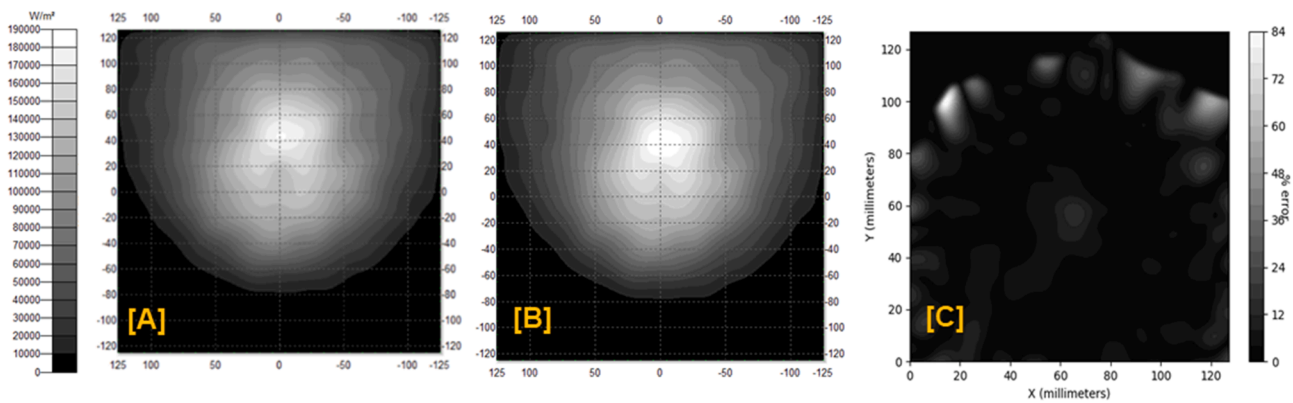


Fig. 8. Comparison of flux maps (with an arbitrary selected source power of 6 kW) generated using optical modelling A) at the original coordinates $(-0.76, -7.14, 0.13)$ mm and B) the predicted ones $(-0.59, -6.93, 0.32)$ mm, including C) their % error graph.

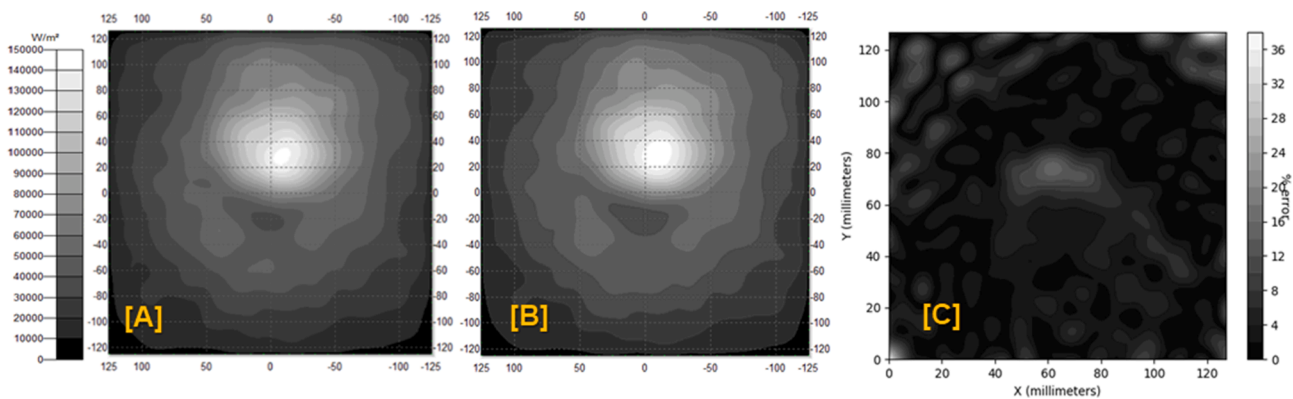


Fig. 9. Comparison of flux maps (with an arbitrary selected source power of 6 kW) generated using optical modelling A) at the original coordinates $(-1.14, -4.94, 6.27)$ mm and B) the predicted ones $(-1.06, -4.90, 6.49)$ mm, including C) their % error graph.

3.3. Fine-Tuning the model

The Gaussian form of the incident flux due to the optical concentration plays a significant role in selecting the training dataset and, consequently, on the CNN model's performance. In the experimental setup, a 3 mm movement around the focal point affected the max flux by approximately 2% and the average flux by approximately 8%. The same movement of 3 mm around a light source already displaced by 10 mm decreased the max flux by more than 36% and the average by more than 37%. Note that the 10 mm displacement would already have decreased the max flux by 86% and the average by 51%. This means that even if the goal is to maintain a high level of accuracy close to the focal point, the training set should include data further from the focal point and certainly up to the selected limits. Fig. 7 clearly showcases the importance of limiting the CNN predictions within the training range. An important note for future model development is to design around the desired and expected boundaries of the system.

Although optical modelling was not discussed in-depth, some parameters indirectly played an important role in the CNN model's development and performance, such as the number of individual rays. A relatively small number of rays ($\sim 50,000$) were chosen to manage the total computational time of the more than 2,500 simulations. A low number of rays in a Monte Carlo ray tracing simulation will result in a higher variation between simulations of the same geometry and parameters, and vice versa. Some of the local errors visualized in Fig. 8 and Fig. 9 can be attributed to the low number of rays, despite the smoothing algorithm applied on every simulated contour flux map. To investigate the effect of the number of rays, an additional 200 simulations were executed using 3x more rays (i.e., 150,000 rays) and added to the

existing dataset. As a result, accuracy improved to 0.07, 0.11, and 0.07 mm in the x-, y-, and z-directions, respectively, with an absolute error of less than $|d'|_{\text{HFSS}} = 0.147$ mm; an improvement of more than 1.7x compared to the original dataset. Notably, the addition of 200 simulations with the initial number of rays (50,000) minimally affected the final model's overall accuracy. This result indicates that the CNN model's accuracy can be improved by improving the numerical data's quality rather than just the training dataset's size.

4. Conclusion

The positioning of the light source (lamp filament) of a light concentrating system is critical, with peak flux and average power reduced significantly for a poorly focused system. The importance of positioning is clearly demonstrated in Fig. 4A (focused system) and Fig. 9A (few millimeters displaced source), where peak flux reduces ~ 2.5 x times and total incident power reduces ~ 1.4 x times. In this context, the study's central hypothesis was that a machine learning model based on CNN (Convolutional Neural Network) could predict the position of the light source of an HFSS (High Flux Solar Simulator) with an accuracy equal to or better than the human operator. The hypothesis was validated as the CNN achieved very similar accuracies (0.163 mm in the x-direction, 0.141 mm in the y-direction, and 0.125 in the z-direction) as that of the mechanical slider controlling the light source position (estimated as 0.14 mm approximately in every axis/direction). These prediction accuracies were improved further with the addition of refined simulations (i.e., 0.07, 0.11, and 0.07 mm, respectively). The CNN training and prediction bias were avoided by utilizing only simulated flux maps (optical modelling).

All in all, one of the prime motivations of this study was to save the operators' time and reduce the effort of this tedious task, given that a trained CNN model can execute the task in quasi-instantaneous time. On the other hand, a trained operator could need upwards of 30 min to position and focus the lamps when starting from a poor initial state, based on the author's experience. This study discussed other equally important findings and challenges:

- Contrary to typical engineering applications, CNN's operational range should remain within the machine learning training data range.
- Similarly, extensive tests support that training data should not be collected following specific input patterns (e.g., fixed steps, Gaussian distribution steps) to ensure the algorithm does not learn and recognize these input patterns.
- Besides the intuitive approach to improving the predictions of CNN modelling by increasing the size of the training dataset (e.g., number of images), a larger improvement can likely be achieved by enhancing the data's quality. This approach was demonstrated by using detailed simulations generated by increasing the number of the total rays (3x) to improve the prediction accuracy (1.7x).
- The combination of simulations and measurements also improves the CNN's predictions, although this effect was not quantified.
- Conversely, the sole use of ideal numerical simulations as training data appears prone to overfitting errors because of the absence of manufacturing imperfections. For instance, due to the symmetry along the z-direction and the generation of extremely similar flux maps, the developed CNN model proved incapable of extracting enough features to distinguish between the positive and negative z-direction. This is a limitation that may affect some HFSS applications more than others, therefore it needs future study.

Many more aspects, approaches, and capabilities can be investigated towards even more sophisticated systems, such as integrating the CNN into a predictive control loop that focuses each light source with no human intervention. Another aspect is the assimilation of more degrees of freedom (DOF) beyond the three-axis presented here, e.g., roll-pitch-tilt angles and the reflector DOF. Similarly, the incorporation of the rest of the light sources of the HFSS with simultaneous operations is an approach worth investigating. Finally, the capability to generate flux maps (irradiances) beyond the usual form (i.e., concentric contours with Gaussian distribution of flux) to power innovative and radical light concentrating technologies is another particularly useful avenue of study for further investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Author Credit Statement

D. Haseler: Data curation, Validation, Visualization, Writing. **A.M. Ali:** Software, Investigation, Formal analysis, Writing. **K.E. Kakosimos:** Conceptualization, Funding acquisition, Supervision, Writing.

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